

New Threshold and Shrinkage Function for ECG Signal Denoising Based on Wavelet Transforms

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Abstract—In order to denoise the noisy ECG signal efficiently and keep the denoised ECG signal in distortion-free and smooth, a new threshold and shrinkage function are proposed in this paper based on the Multi-analysis wavelet threshold denosing. The parameters of the function were optimized by denoising the noisy ECG signal. The denoising effects of different shrinkage function for ECG signal were compared. The results show that the highest SNR can be reached using the proposed function to denoise ECG signal with different levels of contamination.

Keywords—threshold; shrinkage; ECG; denoise;

I. INTRODUCTION

The detection and analysis of electrocardiogram (ECG) are very important in clinical diagnose. It's the primary reference to understand cardiac function and status, diagnosing cardiovascular disease and evaluating the validity of difference kinds of therapies. The process of ECG signal acquisition is non-invasion and easy to operate, but it is also easy induced some noises. The ECG signal combined with these noises will cause misdiagnosis in clinical application. So, performing some denoised processing for the acquired ECG signals so that it can reflect patient's cardiac status exactly is very importance.

The key steps of signal denoising based on wavelet transform are the threshold and shrinkage function selection. The universal threshold, which is proposed by D. L. Donoho[1], is the most popular in all kinds of threshold. It can be expressed as follows

$$T = \sigma \cdot \sqrt{2 \log(N)} \quad \sigma = \frac{\text{Median}(|c|)}{0.6745} \quad (1)$$

Where N is the sampling length of noisy signal, σ is the standard deviation of zero-mean additive white Gaussian noise (WGN) estimated by Donoho and Johnston. c is the detail coefficients of wavelet transform.

After several decades years' development the expressions of shrinkage function have lots of kinds. Such as soft shrinkage function, hard shrinkage function [2], firm shrinkage function [3], hyper-trim shrinkage function [4], multi-parameter best-basis thresholding shrinkage function [5], Yasser shrinkage function [6] et al. All these thresholds and shrinkage functions promoted the application of wavelets in signal denoising

extremely. In this paper, four kinds of shrinkage functions were introduced and compared with the proposed one.

In 1995 Donoho and Johnston developed the hard and the soft shrinkage functions were expressed as formula (2) and (3) respectively.

$$\hat{Y} = \begin{cases} 0 & |Y| \leq T \\ Y & |Y| > T \end{cases} \quad (2) \quad \hat{Y} = \begin{cases} 0 & |Y| \leq T \\ Y - T & Y > T \\ Y + T & Y < -T \end{cases} \quad (3)$$

The derivation of standard soft shrinkage function is not continuous. Both hard and soft shrinkages have advantages and disadvantages. The soft shrinkage estimates tend to have a bigger bias, due to the shrinkage of large coefficients. Due to the discontinuities of the shrinkage function, hard shrinkage estimates tend to have a bigger variance. In other words, it will be sensitive to small changes in the signal.

To overcome the drawbacks of hard and soft shrinkage, a firm shrinkage function was introduced by Gao and Bruce,

$$\hat{Y} = \begin{cases} 0 & |Y| \leq T_1 \\ \text{sgn}(Y) \left[\frac{T_2(|Y| - T_1)}{(T_2 - T_1)} \right] & T_1 < |Y| \leq T_2 \\ Y & |Y| > T_2 \end{cases} \quad (4)$$

Where, T_2 is decided by formula (1), the scope of T_1 is $0 \sim T_2$. According to the previous experiments when T_1 equals $2/3 T_2$, the denoised results would be better. The shrinkage function is continuous and approaches the identity line as $|Y|$ increase. The firm shrinkage function provides a good compromise between the hard and the soft shrinkage function. The firm shrinkage is less sensitive than hard shrinkage to small fluctuations and less biased than soft shrinkage.

In 2006 Yasser presented an improved threshold shrinkage function as formula (5).(named Yasser shrinkage function)

$$\hat{Y} = \begin{cases} Y & |Y| \geq T \\ \text{sgn}(Y) \cdot \frac{|Y|^\gamma}{T^{\gamma-1}} & |Y| < T \end{cases} \quad (5)$$

Where, the value of T is decided by formula (1). The Yasser shrinkage just shrinks the wavelets coefficients which are lower than threshold and it keep the continuing of function.

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In the reference [6], $\gamma=3$ was used to finish the speech signals denoising and obtained good results.

The shrinkage expressed in formula (4) and (5), combined the advantages of hard shrinkage with soft shrinkage, possess better results in signal denoising. But it wasn't designed for ECG signal specially and could not obtain the best result in denoising processing. For firm shrinkage, the denoised result of ECG signal is closely to hard shrinkage when contamination is slight. For Yasser shrinkage, the denoised signal can't be smooth enough when contamination is serious.

A new shrinkage function was proposed in this paper. The parameters of this shrinkage function were optimized by comparing denoised results of the simulative ECG signal at different contaminating levels. In order to verify the denoised results of the presented shrinkage conveniently, the symlets wavelet (sym6), 5-level and global thresholding were used for the whole denoising process.

II. THRESHOLD AND SHRINKAGE FUNCTION

A. shrinkage function proposing

The choice of threshold and shrinkage function is the most important step for wavelet denoising. In order to obtain the best denoising results, a new shrinkage function which would be used in ECG signals denoising was proposed here, expressed as formula (6).

$$\hat{Y} = \begin{cases} 0 & |Y| \leq T_L \\ \text{sgn}(Y) \left[\frac{|Y - T_L|^\gamma \cdot T_H}{|T_H - T_L|^\gamma} \right] & T_L < |Y| \leq T_H \\ Y & |Y| > T_H \end{cases} \quad (6)$$

Where γ , T_H , T_L are alterable. Formula (6) is equal to formula (2) when $T_L = T_H$; formula (6) is equal to formula (4) when $\gamma=1$, $T_L=2/3T_H$; and formula (6) is equal to formula (5) when $\gamma=3$, $T_L=0$. Obviously, the shrinkage function expressed in formula (6) synthesized many formats. The following work is to figure out a best format which can produce the best denoising result for ECG signal.

B. Simulative ECG signal building up

To decide the parameters of the proposed shrinkage function, a standard ECG signal should be built up first, as shown in Fig. 1

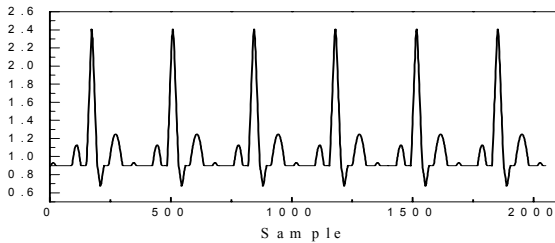


Figure 1. Simulative ECG Signal

The ECG signal containing 2048 sampling points. P wave, QRS wave, ST wave, T wave and U wave all have been

described precisely. The amplitude and frequency of simulative ECG signal are similar to the real one. Generally the clinical ECG signals always contain noise (however the contamination is slight). The noisy ECG signal used in experiment was realized by superposing WGN on simulative ECG signal.

C. Parameters selection

Considering different contamination levels and making the pending signal more similar to the real signal, the variable range of WGN was set from 0~0.2. The range of γ is 0~10, the range of T_H is 0~2, the range of T_L is 0~ T_H . The denoising process was performed in terms of formula (6). The signal to noise ratio (SNR) of denoised ECG signals was calculated and a four-dimension data was obtained. Where, the calculation formula of SNR is expressed as follows

$$SNR = 10 \times \log_{10} \left(\frac{\sum_{n=1}^N s^2(n)}{\sum_{n=1}^N |s(n) - \hat{s}(n)|^2} \right) \quad (7)$$

Where, $s(n)$ is the sampling value of simulative ECG signal at the n^{th} point, $\hat{s}(n)$ is the sampling value of denoised ECG signal at the same point.

1) Value selection of γ

First select 20 different values for the variable of WGN to set up 20 groups pending ECG signals as the experimental signal. Second take different γ value into the proposed shrinkage function to denoise. Third calculate the sum of SNR for 20 groups denoised signals after denoising by different γ value. γ value as a function of ΣSNR was shown in Fig. 2. According to Fig. 2 when $\gamma=3.5$, ΣSNR will reach to the maximum which is 479.89. The higher SNR is the better denoising result can be obtained. So γ is 3.5.

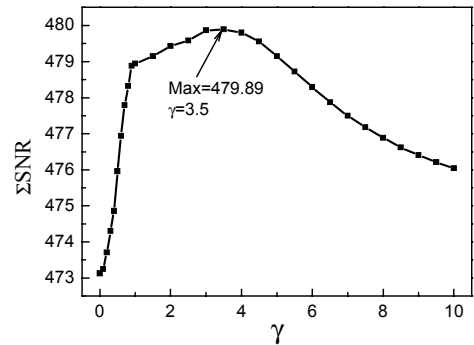


Figure 2. Relationship between γ and ΣSNR

2) Value of T_H

When $\gamma=3.5$ and according to the maximum SNRs of each signal, the relationship between T_H and σ was shown in Fig. 3. The experimental results were fitted by a solid line. The dash line results from formula (1). T_H which corresponded to maximal SNR increased linearly with variable.

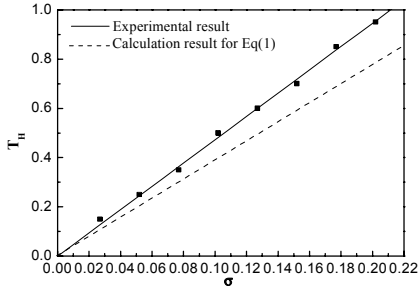


Figure 3. Relationship between T_H and σ

According to the fitting result, T_H can be written as follows,

$$T_H = 1.2\sigma \cdot \sqrt{2\log(N)} \quad (8)$$

3) Value of T_L

Assume σ and T_L as variables, the calculation results of SNR was meshed in Fig. 4. According to Fig.4 (a) when σ changed from 0-0.2, a set of maximum SNRs can be obtained. Corresponding to these maximum SNR also exist a set of T_L value. The relationship between σ and T_L was shown in Fig. 4(b). Triangle symbol represents the maximum SNRs. The solid line is the fitting result. The dash line shows the calculation result of formula (1). According to Fig. 4 T_L can be expressed as follows,

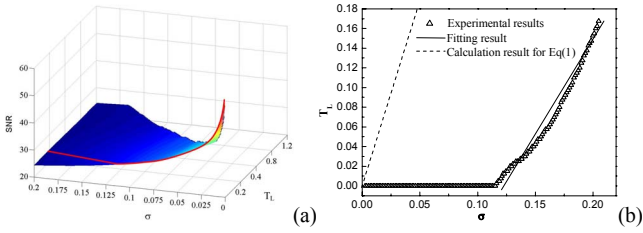


Figure 4. Relationship in T_L , σ and SNR. (a) T_L , σ and SNR; (b) T_L and σ

$$\begin{cases} T_L = 0 & \sigma \leq 0.121 \\ T_L = 0.4865(\sigma - 0.121) \cdot \sqrt{2\log(N)} & \sigma > 0.121 \end{cases} \quad (9)$$

Thus, the parameters (γ , T_H and T_L) of formula (5) are decided as follows,

$$\begin{aligned} \gamma &= 3.5, \quad T_H = 1.2\sigma \sqrt{2\log(N)} \\ \begin{cases} T_L = 0 & \sigma \leq 0.121 \\ T_L = 0.4865(\sigma - 0.121) \cdot \sqrt{2\log(N)} & \sigma > 0.121 \end{cases} \end{aligned}$$

III. RESULTS AND DISCUSSION

Wavelets denoising effect which is good or not, mainly decided by the similarity between original signal and denoised signal and the smoothness of denoised signal. SNR and root of mean square error (RMSE) can reflect the effect directly. And the RMSE can be denoted by formula (10).

$$RMSE = \sqrt{\frac{1}{N} \sum_{n=1}^N (s(n) - \hat{s}(n))^2} \quad (10)$$

where, $s(n)$ is the sampling value at the n^{th} point of the original signal, $\hat{s}(n)$ is the sampling value at the n^{th} point of the denoised signal.

A. Simulation

The simulation was carried out by MATLAB. WGN with different noise levels has been added to the simulative ECG signal to test the denoising efficiency. The soft, hard, firm, Yasser and the presented shrinkage were used to carry out the simulation. The SNRs and RMSEs of denoised ECG signals, which denoised by the method mentioned above, were calculated respectively, as shown in Fig. 5.

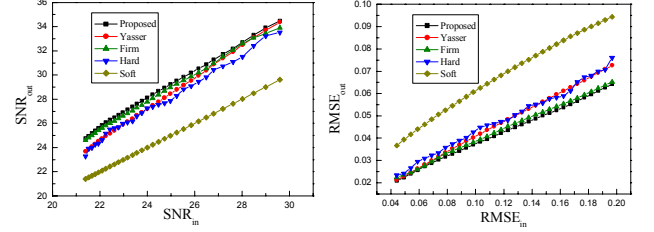
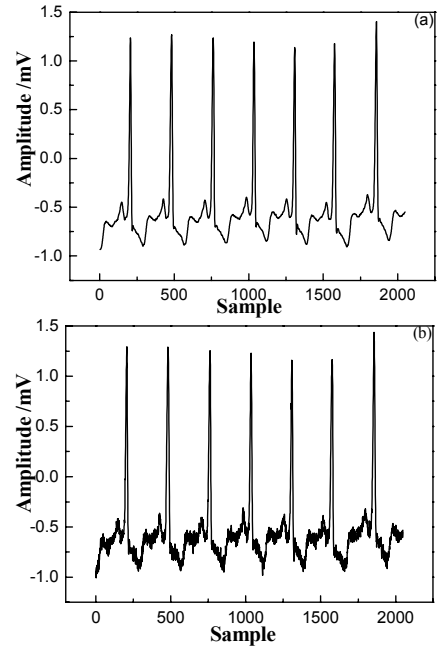


Figure 5. Relationships between SNR_{in} and SNR_{out} (a) $RMSE_{in}$ and $RMSE_{out}$ (b)

It is clear from the Fig. 5 that the proposed method can reach the best result and better than other methods. Whichever levels of contamination the proposed method possesses the maximal SNR and minima RMSE.

B. Real ECG signal denoising

After denoising and smoothing, record 223 in MIT-BIH arrhythmia database was regarded as the original ECG signal and shown in Fig. 6(a). The pending signal shown in Fig. 6(b) was realized by superposing WGN (with variance 0.04) on original signal. Aim of the following experiments is to compare the denoised efficiency of the proposed shrinkage, Yasser shrinkage, firm shrinkage, hard shrinkage and soft shrinkage. The symlets wavelet (sym6) and 5-level were also used in all the experiments. Fig. 6 (c-g) was the denoised results.



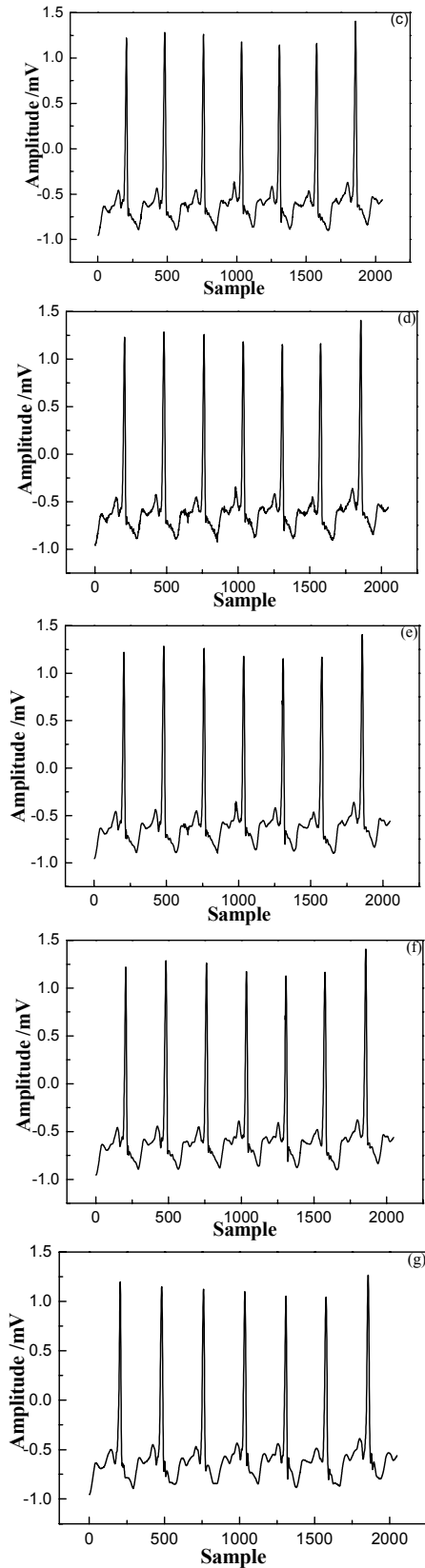


Figure 6. Schematic of ECG signal and processed results. (a) original signal; (b) noisy signal; (c) denoised signal by the presented shrinkage; (d) denoised signal by Yasser shrinkage; (e) denoised signal by firm shrinkage; (f) denoised signal by hard shrinkage; (g) denoised signal by soft shrinkage

Fig. 6 (a-g) are the original signal, noisy signal, denoised signal by different shrinkage respectively. The denoised results were listed in Tab. 1.

TABLE 1 DENOISED RESULTS OF NOISY SIGNAL ($SNR_{in}=25.105$, $RMSE_{in}=0.036995$)

Shrinkage	soft	hard	firm	yasser	presented
SNR	25.327	32.092	32.757	32.779	33.038
RMSE	0.036058	0.016549	0.01533	0.01529	0.014841

Tab. 1 shows that both SNR and RMSE of proposed method are better than the other four methods due to good similarity and smoothness.

IV. CONCLUSION

A new threshold and shrinkage function were proposed in this paper. The parameter value for ECG signal was optimized by denoising experiments. The pending signal was realized by superposing WGN on the simulative ECG signal and the real ECG signal obtained from MIT-BIH arrhythmia database. The denoising effects for noisy ECG signal using different functions were compared here. The results show that the shrinkage function proposed in this paper is very good at noisy ECG signal denoising. Not only can it get the highest SNR, but also keep the similarity and smoothness of denoised signal. In fact, this function also can be used for all kinds of signals denoised.

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